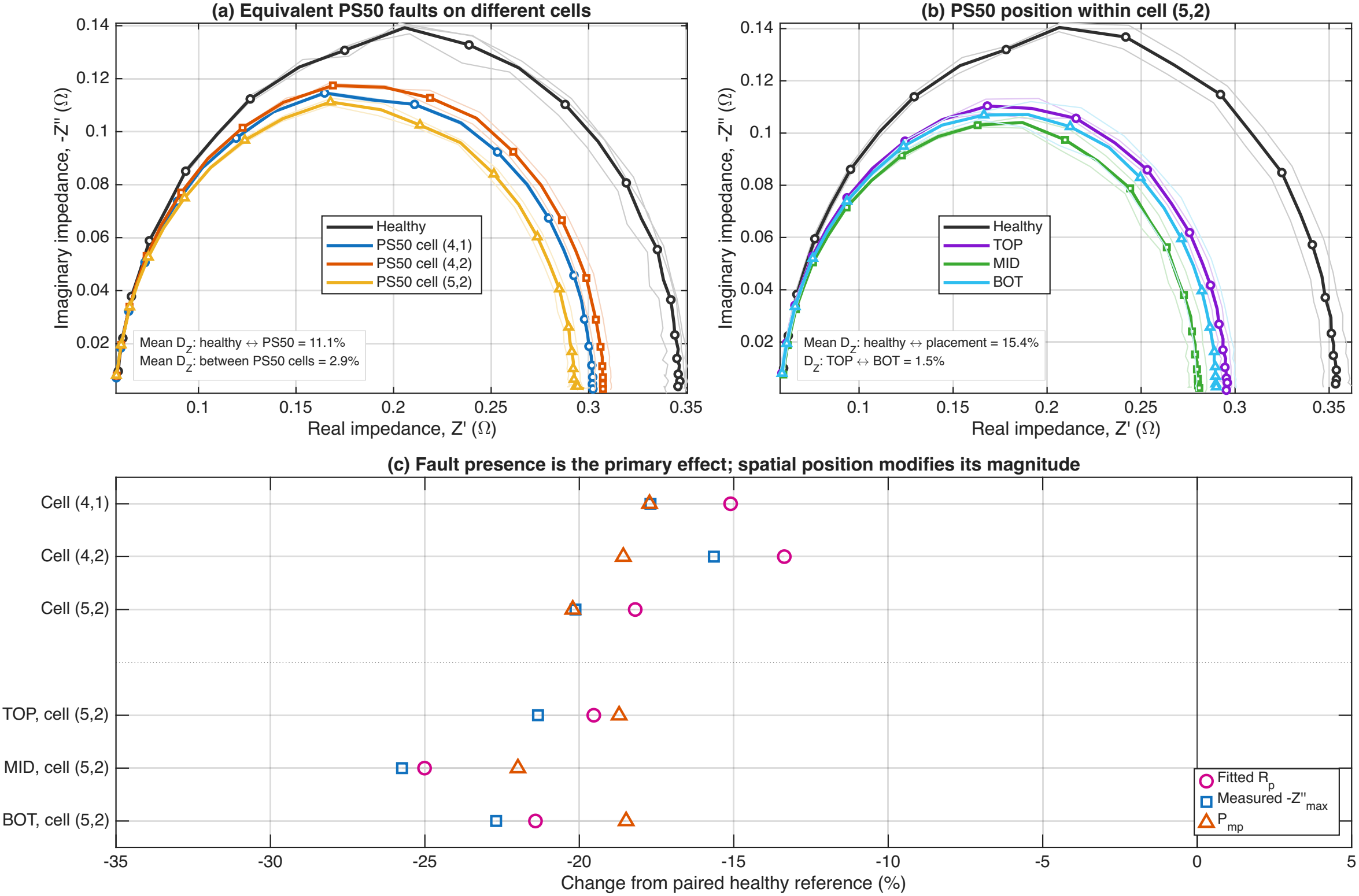

(a) Equivalent PS50 faults on different cells
Imaginary impedance, -Z'' (Ω)
Real impedance, Z' (Ω)
Healthy
PS50 cell (4,1)
PS50 cell (4,2)
PS50 cell (5,2)
Mean D_Z: healthy ↔ PS50 = 11.1%
Mean D_Z: between PS50 cells = 2.9%
(b) PS50 position within cell (5,2)
Imaginary impedance, -Z'' (Ω)
Real impedance, Z' (Ω)
Healthy
TOP
MID
BOT
Mean D_Z: healthy ↔ placement = 15.4%
D_Z: TOP ↔ BOT = 1.5%
(c) Fault presence is the primary effect; spatial position modifies its magnitude
Cell (4,1)
Cell (4,2)
Cell (5,2)
TOP, cell (5,2)
MID, cell (5,2)
BOT, cell (5,2)
Fitted R_p
Measured -Z''_max
P_mp
Change from paired healthy reference (%)